

FTR

Core Formalization

Future trajectory representation for autonomous-agent runtimes:
possible futures, disagreement, future information requirements, proof obligations, risk, resolution and calibration.

$$\mathcal{F}_t = (P_t(\tau), D_t, \mathcal{H}_t, \Omega_t^*, P_t^{\text{proof}}, R_t, \Gamma_t)$$

Status. FTR is **not a validated theory**. Everything here marked **OPEN** is a hypothesis or a candidate formalization; nothing in this document is implemented as FTR in MØBIUS, and no FTR experiment has yet produced a result. Known abstractions — probabilistic forecasting, runtime verification, provenance, value of information — are not relabelled here as FTR novelty.

Version	2026-09-19 · supersedes the undated working specification
Evidence base	MØBIUS repository at <code>61f7100</code> (R3 handoff), evidence directories <code>r1/f9b27dd</code> , <code>r2/f9b27dd</code> , <code>p4-expectations/ba4f291</code> ; research-lead records of 18–19 Sep 2026
Companion post	marcyy.me/posts/mobius-a-correct-world-is-not-a-proven-one
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0 • What changed in this version

- **Evidence horizon, corrected.** The previous version implied information that becomes *unrecoverable* after an action. In every R3 P3 journal with a transition-shaped criterion, the pre-state *was* captured before the first effect; it was absent from the judge's *view*, not from the journal. The unrecoverable variant has not been observed and is kept as a hypothesis (§6).
- **Transition witness, downgraded.** Retired as a new primitive (before-image + provenance edge + past-time operator). Kept as a candidate *representation* (§7).
- **Measured additions.** Hand-coded cause of all 74 `cannot-tell` verdicts; a manual audit of all 20 `established` verdicts (one unsupported); certificate closure measured at 0 of 11 (§12).
- **Vacuity made explicit.** R3 P3's "0 false completions" had zero opportunities to fail (39/39 rows passed their checker) and is recorded as not evaluable (§15).
- **Write-set anticipation.** Stated precisely: a kill threshold fixed before the computation, on an experiment-arm corpus, not a workload (§15).

1 • Motivation — what the evidence points at

MØBIUS's completion plane was the place its questions moved in September 2026. A separate model call (the *judge*, ADR 0043) decides whether a criterion no assertion checks is `established`, `not-established` or `cannot-tell`, from the observations it is shown (ADR 0045: the binding's evidence plus the run's latest non-stale observation of each other resource). In R3 P3 (`claude-opus-5`, three repetitions, 39 valid rows) the judge ruled 114 times: 20 / 20 / 74.

what a <code>cannot-tell</code> reason says is missing	n
prior state only	40
prior state and an unread current resource	5
an unread current resource only	28
judge call timed out	1

The 45 reasons that cite a missing past sit in three unrelated task shapes — rename (10/10), edit-to-spec with "nothing else changed" (20/22), split a document (15/27) — and in none of the other three tasks (0/15). Their criteria are predicates over a *transition*, not a state:

$$C \neq f(W_{\text{post}}), \quad C = f(W_{\text{pre}}, A, W_{\text{post}})$$

FTR's surviving question follows from this: **can a runtime know, before it acts, what a future decision or a future proof will need it to have kept?**

2 • Runtime state and history

$$X_t = (W_t, H_t, G_t, M_t, C_t, E_t, \Gamma_t)$$

$$H_t = (O_0, A_0, E_1, O_1, A_1, E_2, \dots, A_{t-1}, E_t, O_t)$$

History is not state. A current state is a function of history, and not an injective one:

$$H_t \not\equiv W_t, \quad W_t(h_i) = W_t(h_j) \not\Rightarrow h_i = h_j$$

This matters whenever a criterion concerns preservation, movement, non-modification, provenance or causality — any past-to-present relation.

3 • Possible futures as the object

$$\tau_{t:t+k} = (S_t, A_t, E_{t+1}, O_{t+1}, S_{t+1}, \dots, A_{t+k-1}, E_{t+k}, O_{t+k}, S_{t+k})$$

$$P_t(\tau) = P_\theta(\tau_{t:t+k} \mid H_t, W_t, G_t, C_t, M_t), \quad \mathcal{T}_t = \{(\tau_i, p_i)\}_{i=1}^N, \quad p_i \geq 0, \quad \sum_{i=1}^N p_i = 1$$

The target is a distribution over structured trajectories — actions, effects, observations, failures, retries, human intervention, evidence capture, verification, completion — and never "the next action".

4 • FTR is not a plan

$$\mathcal{F}_t \neq \text{Plan}_t, \quad P(\tau \mid X_t, \text{Plan}_t) \neq \mathbf{1}[\tau = \text{Plan}_t]$$

A plan is prescriptive: *what do I intend to do?* FTR is anticipatory: *which futures remain reachable from here, where do they diverge, what will they require, and what should the runtime preserve or inspect now because of that?* It must admit futures in which the plan fails, branches, stalls on a person, loses its evidence or is replaced.

Kill condition: an FTR implementation that reduces to the planner under another name.

5 • Future disagreement and uncertainty

Let $z \in Z$ be equivalence classes of materially distinct outcomes or trajectory shapes.

$$D_t = 1 - \sum_{z \in Z} P(z \mid X_t)^2, \quad \mathcal{H}_t = - \sum_{i=1}^N p_i \log p_i$$

D_t is the Gini–Simpson index over outcome classes: $D_t \approx 0$ when rollouts agree, $D_t \rightarrow 1$ when they scatter. It does not require knowing which future is correct. **PRIOR ART** Sample disagreement as an uncertainty estimate: self-consistency (Wang et al., ICLR 2023), semantic entropy over equivalence classes (Kuhn, Gal, Farquhar, ICLR 2023), conformal ask-for-help sets (Ren et al., CoRL 2023). **OPEN** Whether divergence is a useful *runtime trigger* — for observation, preservation, a question to a person, or delaying an irreversible effect.

6 • Future information requirements and proof obligations

Let $E_\Omega(h)$ be the evidence history h exposes when the fact set Ω is captured, and $C(h) \in \{0, 1\}$ whether the criterion holds in h . The cheapest sufficient capture:

$$\Omega_t^* = \arg \min_{\Omega} \text{Cost}(\Omega) \quad \text{s.t.} \quad \forall h_i, h_j : E_\Omega(h_i) = E_\Omega(h_j) \Rightarrow C(h_i) = C(h_j)$$

Equivalently $C(h_i) \neq C(h_j) \Rightarrow E_\Omega(h_i) \neq E_\Omega(h_j)$: no two possible histories may look identical under the evidence while disagreeing on the criterion. This is the indistinguishability test for evidential soundness. For the rename, $\Omega_t^* = \{\text{digest}(\text{notes}_{\text{pre}})\}$.

$$P_t^{\text{proof}}(a) = P(\neg \text{Provable}(C_{t+k}) \mid a, X_t) \quad \text{is not} \quad P(\text{TaskFailure} \mid a)$$

$$\Delta_{\text{preserve}} = P_t^{\text{proof}}(a) - P_t^{\text{proof}}(a \mid \text{preserve}(\Omega)), \quad \text{preserve}(\Omega) \text{ worthwhile iff } \Delta_{\text{preserve}} \cdot V_{\text{proof}} > \text{Cost}(\Omega)$$

An action may be operationally safe and epistemically dangerous: $P(\text{succeeds} \mid a) \approx 1$ while $P(\text{Provable}(C) \mid a) \ll 1$. MEASURED T4 is one real instance of the second quantity collapsing *for the judge's view* after a move, although the journal still held the proof. OPEN Whether Ω^* or P^{proof} can be estimated before an action, for criteria written in natural language. OPEN The *unrecoverable* case — a fact that exists only before the action — has a minimal construction (move with no prior read) and no observed instance.

7 • Transition witnesses

$$\omega_t = (\phi_{\text{pre}}(W_t), a_t, \phi_{\text{post}}(W_{t+1}), \rho(W_t, a_t, W_{t+1})), \quad \mathcal{T}_{t \rightarrow t+1} = (W_t, A_t, W_{t+1}, R_t)$$

$$\rho_{\text{move}} : \text{digest}(\text{src}@t) = \text{digest}(\text{dst}@t+1) \wedge \text{exists}(\text{src}@t+1) = 0 \wedge \text{exists}(\text{dst}@t+1) = 1 \Rightarrow \omega_t \models C$$

RETIRED as a new primitive: it is a before-image, a provenance edge and a past-time temporal operator (Havelund & Roşu, TACAS 2002; Snodgrass & Ahn 1986; Buneman et al. 2001; Green et al. 2007). CANDIDATE as a representation. OPEN The systems question is narrower: which transitions an agent runtime must make first-class evidence, and how it decides that *before* acting. A digest shows equal bytes, not the causal path; for “moved, not rewritten” it is insufficient.

8 • Risk over possible futures

$$R_t(a) = \underbrace{\mathbb{E}_{\tau \sim P_t}[L(\tau, a)]}_{\text{bad outcome}} + \underbrace{\lambda_D D_t(a)}_{\text{divergence}} + \underbrace{\lambda_P P_t^{\text{proof}}(a)}_{\text{unprovable completion}}$$

9 • Runtime metareasoning

$$\mathcal{X}_t = \{\text{act}, \text{observe}, \text{test}, \text{ask}, \text{fork}, \text{replan}, \text{preserve}, \text{wait}\}$$

$$V(\mathcal{F}_t) = -[\alpha \mathcal{H}_t + \beta D_t + \gamma R_t + \delta P_t^{\text{proof}}]$$

$$x_t^* = \arg \max_{x \in \mathcal{X}_t} \left[\mathbb{E}[V(\mathcal{F}_{t+1}) \mid x, X_t] - V(\mathcal{F}_t) - \lambda_C \text{Cost}(x) - \lambda_L \text{Latency}(x) \right]$$

PRIOR ART Value of information (Howard 1966) and rational metareasoning (Russell & Wefald 1991) supply the decision rule; speculative execution (Nightingale et al., SOSp 2005) is the systems precedent for acting on a predicted future. The only candidate contribution is the *state* the rule is computed over — in particular *preserve*, choosing what to keep because of what a future proof will need. Its cheapest rival is a wholesale snapshot, which on a filesystem costs almost nothing.

10 • Forecast identity, resolution and calibration

$$f_i = (W_i, B_i, \tau_i, p_i, \Omega_i, T_i), \quad \text{Valid}(f_i, t) \iff B_i \simeq B_t \wedge W_i \simeq W_t$$

W_i	world version the forecast was made against
B_i	basis: observations, evidence, assumptions, runtime state
τ_i	forecast trajectory or outcome class
p_i	probability / confidence
Ω_i	predicted future information requirement
T_i	resolution condition / horizon

$$\text{Resolve}(f_i, O_{i+1:t}) \in \{\text{satisfied}, \text{violated}, \text{void}\}$$

satisfied — the basis held and the predicted event occurred. **violated** — the basis held and it did not. **void** — the world or premise changed so the forecast no longer applies; not a prediction error. The same version-binding appears in MØBIUS's shipped approval revalidation (`reusable` | `superseded` | `unevaluable`) and in published commit-time authorization (CommitGuard, arXiv 2607.10487) — it is `PRIOR ART` as a pattern.

$$y_i = \mathbf{1}[\text{Resolve}(f_i) = \text{satisfied}], \quad \text{BS} = \frac{\sum_{i: r_i \neq \text{void}} (p_i - y_i)^2}{|\{i : r_i \neq \text{void}\}|}$$

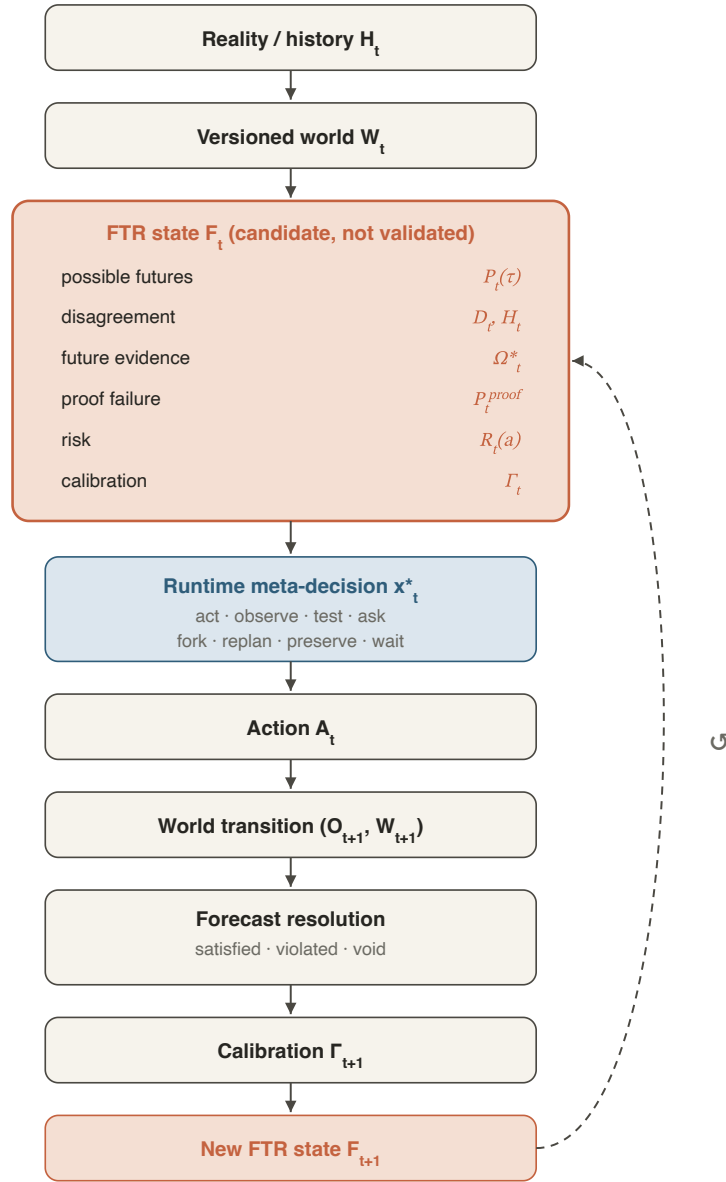
$$\Gamma_t = \text{CalibrationHistory}(f_1, \dots, f_t), \quad \text{Trust}_{t+1} = g(\Gamma_t)$$

`NOT EVALUABLE` on existing data. R3 P4 resolved 1,345 model-written `expected` assertions from 104 journals: they are bare Booleans. A constant forecast has one bin, so the Murphy resolution term is zero by definition whatever the outcomes, and a kill criterion "resolution ≈ 0 " fires on arithmetic. Calibration is a *requirement* on any future FTR experiment, not a result.

11 • The candidate object and its loop

$$\mathcal{F}_t = P_\theta(\tau_{t:t+k}, \Omega_{t:t+k}, R_{t:t+k} \mid H_t, W_t, G_t, C_t, M_t)$$

$$X_t \rightarrow \mathcal{F}_t \rightarrow x_t^* \rightarrow A_t \rightarrow (O_{t+1}, W_{t+1}) \rightarrow \text{Resolve} \rightarrow \Gamma_{t+1} \rightarrow \mathcal{F}_{t+1}$$



Only the Boolean expectation layer exists in code today; nothing here is implemented as FTR.

Figure 1. FTR as a candidate closed loop. Only a Boolean expectation layer exists in code.

12 · Completion soundness and certificate closure

Let J_i be the evidence the judge was actually shown for criterion i and C_i the evidence the final completion record keeps.

closure: $J_i \subseteq C_i$

replay: $\text{Judge}(\text{Criterion}_i, C_i) = \text{OriginalDecision}_i$

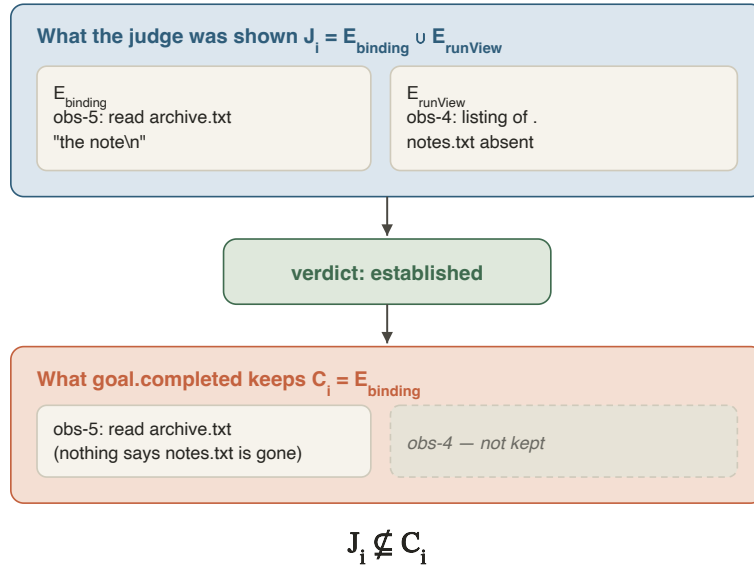
soundness: $\text{Judge}(C, E) = \text{established} \Rightarrow E \models C$

world correctness $\not\equiv$ evidential correctness

MEASURED In MØBIUS at `f9b27dd` the judge is shown $J_i = E_{\text{binding}} \cup E_{\text{runView}}$, while `#bindClaimed` records $C_i = E_{\text{binding}}$ and `goal.completed` carries it. Across R3 P3, 11 judge-bound criteria appear in completed runs;

$J_i \subseteq C_i$ holds in **0**; in 8 the omitted evidence carries part of what the judge's own reason cites. The journal's `criterion.judged` event still names J_i : the gap is in the certificate, not the log. Replay has not been run.

MEASURED Manual audit of all 20 `established` verdicts: 19 supported by what the judge was shown (3 only on a literal reading); **1 not** — R1 T4 repetition 3, where the judge answered `cannot-tell` correctly, was then shown only a read of `archive.txt`, and answered `established` with a reason that never mentions the former content. One in twenty is an existence proof, not a rate.



R3 P3: 11 judge-bound criteria in completed runs; $J \subseteq C$ in 0 of 11.
 In 8 of the 11 the omitted evidence carries part of what the judge's own reason cites.
 The journal's `criterion.judged` event still names J — the gap is in the certificate, not in the log.

Figure 2. The judge's basis against the completion record, T4 repetition 3.

13 · A worked temporal example

Criterion C : *archive.txt exists with the former content of notes.txt, and notes.txt no longer exists.*

$$E_{\text{post}} = \{ \text{archive}_{t+1} = \text{"the note"}, \neg \text{exists}(\text{notes}_{t+1}) \}$$

$$h_1 : \text{notes}_t = \text{"the note"} \rightarrow \text{archive}_{t+1} = \text{"the note"} \quad C(h_1) = 1$$

$$h_2 : \text{notes}_t = \text{"other"} \rightarrow \text{archive}_{t+1} = \text{"the note"} \quad C(h_2) = 0$$

$$E_{\text{post}}(h_1) = E_{\text{post}}(h_2) \Rightarrow E_{\text{post}} \not\models C$$

$$\Omega = \{ \text{digest}(\text{notes}_t) \} : \text{digest}(\text{notes}_t) = \text{digest}(\text{archive}_{t+1}) \wedge \neg \text{exists}(\text{notes}_{t+1}) \Rightarrow C$$

In the real journal (r1/f9b27dd/P3-control-b/T4-rename/rep-3) both digests are `sha256:6862e454af825d8f...`: event 7, before the move, and event 17, after it.

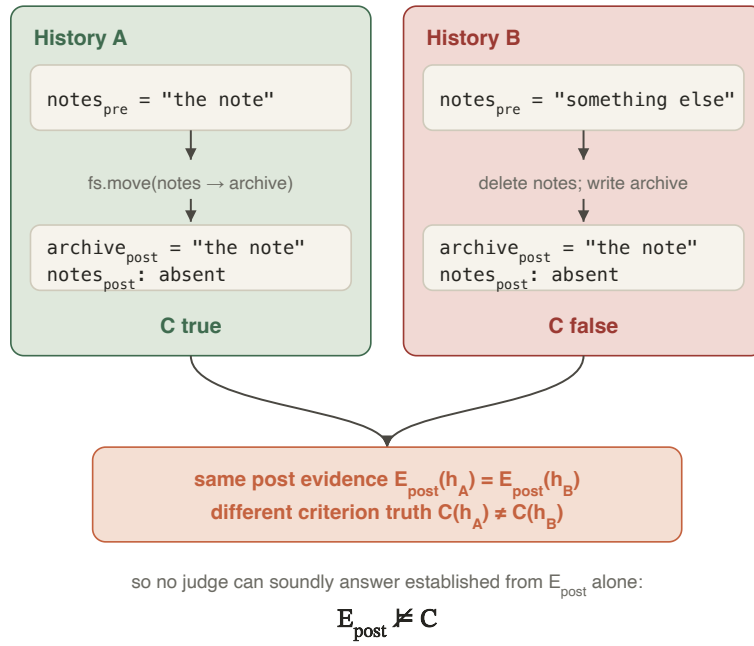


Figure 3. Same post-state evidence, opposite criterion truth.

14 • Notation

W_t	versioned external world state at t
H_t	execution history: observations, actions, effects
G_t, C_t	active goals; completion criteria
M_t	agent / runtime internal state: plan, tasks, subagents, tool context
E_t	evidence currently available (to whom — journal, proposer, judge — must be stated)
Γ_t	forecast resolution and calibration history
$\tau, P_t(\tau), \mathcal{T}_t$	trajectory; distribution over trajectories; sampled approximation
$D_t, \mathcal{H}_t$	disagreement (Gini–Simpson over outcome classes); entropy
Ω_t^*	cheapest fact set that keeps criterion-distinct histories distinguishable
P_t^{proof}	probability that completion becomes unprovable
$R_t(a)$	risk of action a over futures
$\mathcal{X}_t, x_t^*$	meta-action set; chosen meta-action
$f_i, \text{Resolve}, \text{BS}$	forecast instance; resolution; Brier score over non-void forecasts
J_i, C_i	evidence the judge was shown; evidence the completion record keeps

15 · Research status, 19 September 2026

claim / component	status	basis
FTR overall	OPEN	reopened by the author as a question after a recommendation to retire it; no measurement yet
future prediction itself as novelty	RETIRED	predictive runtime verification (Lindemann et al., arXiv 2211.01539)
exact prediction of future actions	UNSUPPORTED	not the surviving claim
read-only pre-write write-set anticipation	KILLED	median read coverage 0.25 (33 of 69 zero) against a recall < 0.5 threshold fixed before computing; experiment-arm corpus, so only the shape split (edit vs create/rename) generalises
Boolean <code>expected</code> → Brier / resolution	NOT EVALUABLE	constant forecasts; resolution zero by definition
rollout disagreement as runtime signal	OPEN	as uncertainty: prior art; as trigger: untested
future information requirement Ω^*	OPEN	formalised; no estimator
anticipatory evidence preservation	OPEN	experiment E4
proof-failure prediction	OPEN	one instance of the quantity; no predictor
transition witness	CANDIDATE	retired as a primitive; kept as representation
evidence horizon (past out of judge's view)	MEASURED OPEN	45 / 74 cannot-tell; three task shapes; general version open
unsupported establishment	MEASURED OPEN	1 clear case in 20; rate unknown (E1)
certificate closure / replayability	MEASURED OPEN	0 / 11 closed; replay not run (E3)
version-bound decisions and forecasts	PRIOR ART	implemented and measured in MØBIUS; published (CommitGuard, PlanFence)
value of information / computation	PRIOR ART	decision machinery, not FTR novelty
three-valued outcomes / partial observation	PRIOR ART	not a novelty claim
"0 false completions" (R3 P3)	NOT EVALUABLE	0 opportunities: 39 / 39 valid rows passed their checker

16 · Prior-art boundary

Before any FTR component is claimed as new it must be separated from: predictive runtime verification and conformal prediction for STL (arXiv 2211.01539); past-time LTL monitoring (TACAS 2002) and runtime verification generally (Leucker & Schallhart, JLAP 2009); temporal databases (Snodgrass & Ahn 1986); provenance

(ICDT 2001; PODS 2007); proof-carrying code (POPL 1997); ATMS (de Kleer 1986); value of information (Howard 1966) and rational metareasoning (Russell & Wefald 1991); speculative execution (SOSP 2005); sampling-based uncertainty (arXiv 2203.11171, 2302.09664, 2307.01928); commit-time and dependency-scoped validation for LLM agents (arXiv 2607.10487, 2609.03340) and read-set reconstruction (arXiv 2605.17076); false success in LLM agents (arXiv 2606.09863).

17 • Candidate invariants

1. A forecast carries explicit W, B, T, p .
2. Invalidation of a forecast's basis resolves it as *void*, never as an ordinary error.
3. Before a world-changing action, evidence-loss and proof-loss are evaluated.
4. Preserved evidence is attributable to a pre-action world version.
5. **established** $\Rightarrow E \models C$ for the evidence actually shown.
6. The completion record contains the decision basis: $J_i \subseteq C_i$.
7. A forecast's influence on action is weighted by measured calibration, never by its rhetoric.
8. Every existence claim is reported with its opportunity count; zero opportunities is "not tested".

18 • Open experiments

E1 • Evidential soundness replay. Freeze criterion, prompt, evidence and model for T4 post-only evidence ($E \not\models C$); re-ask N times.

$$\hat{p}_{\text{est}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\text{Judge}_i(C, E) = \text{established}]$$

$\hat{p}_{\text{est}} > 0$ with $E \not\models C$: unsupported establishment is a property of the judge under that evidence. $\hat{p}_{\text{est}} = 0$: repetition 3 is not reproducible from its inputs.

E2 • Evidence-horizon intervention.

$$A_0 = E_{\text{post}}, \quad A_1 = E_{\text{post}} \cup E_{\text{pre}}, \quad A_2 = E_{\text{post}} \cup \omega_{\text{transition}}, \quad A_3 = E_{\text{post}} \cup \text{proposer self-claim}$$

Measure $P(\text{established} \mid A_k)$, $P(\text{cannot-tell} \mid A_k)$, $P(\text{unsupported} \mid A_k)$. Hypothesis: $A_1, A_2 \gg A_0$ for justified establishment and $A_3 \approx A_0$. If A_3 moves the judge like A_1 , the judge reads assertions rather than evidence.

E3 • Certificate closure audit and replay. $\chi_i = \mathbf{1}[J_i \subseteq C_i]$, $\text{ClosureRate} = \frac{1}{N} \sum_i \chi_i$ (0/11 today); re-judge from C_i alone before any fix, so the fix has a baseline.

E4 • Anticipatory evidence preservation. Before a world-changing action, offer *preserve* or *proceed*; trigger $\mathbf{1}[P_t^{\text{proof}}(a) > \eta]$. Compare with preserve-nothing and preserve-everything (snapshot) baselines on tasks where a later criterion depends on pre-state and matched tasks where it does not:

$$\Delta CT = CT_{\text{base}} - CT_{\text{FTR}}, \quad \Delta UE = UE_{\text{base}} - UE_{\text{FTR}}, \quad \Delta \text{Cost} = \text{Cost}_{\text{FTR}} - \text{Cost}_{\text{base}}$$

Kill: no reduction in cannot-tell or unsupported establishment beyond the snapshot baseline at comparable cost, or no generalisation beyond the three task shapes. Every arm reports its opportunity count.

19 · Evidence index

R2 pilot (0/6 checker, 5/6 declared complete)	<code>evidence/r2/de5b79b</code>
ADR 0043 re-run (false 0/6, runtime 0/6; R1 6/7)	<code>evidence/{r2,r1}/33e599e</code>
R3 P3 (39 valid rows, 114 judgements)	<code>evidence/r2/f9b27dd/{P3-judge-run,P3-judge-run-b}</code> , <code>evidence/r1/f9b27dd/{P3-control,P3-control-b}</code>
R3 P4 (1,345 assertions, 104 journals)	<code>evidence/p4-expectations/ba4f291</code>
layered determinism (0.923 / 0.795 / CV 0.129)	re-derived from R3 P3 journals, 13 tasks × 3
judge and certificate code	<code>packages/runtime-core/src/loop.ts</code> — <code>#bindClaimed</code> , <code>#established</code> , <code>#runView</code>

*What is clearer now is not what the answer is,
but what it is not — and which questions are worth the next experiment.*

Marc Ellington

MØBIUS Research

A handwritten signature in black ink, reading "Marc Ellington". The script is fluid and cursive, with the first letters of "Marc" and "Ellington" being capitalized and prominent.

19 September 2026